

Lakshya JEE 2.0 (2025)

Mathematics

DPP: 4

Indefinite Integration

- Q1** Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.

$$\int \frac{dx}{\sqrt{9-16x^2}}$$

- Q2** Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.

$$\int \frac{dx}{25+4x^2}$$

Q3 $\int \frac{dx}{\sqrt{x^2-2x+3}} =$

- (A) $\log|x-1-\sqrt{x^2-2x+3}|+c$
 (B) $-\log|x-1+\sqrt{x^2-2x+3}|+c$
 (C) $\log|x-1+\sqrt{x^2-2x+3}|+c$
 (D) $\log|x+1+\sqrt{x^2-2x+3}|+c$

Q4 $\int \frac{e^x dx}{\sqrt{1-e^{2x}}} =$

- (A) $\cos^{-1}(e^x) + c$
 (B) $-\cos^{-1}(e^x) + c$
 (C) $\cos^{-1}(e^{2x}) + c$
 (D) $\sqrt{1-e^{2x}} + c$

Q5 $\int \frac{5x+4}{\sqrt{x^2+2x+5}} dx$

- (A) $-5 \cdot \sqrt{x^2+2x+5} - \ln|x+1+\sqrt{x^2+2x+5}| + c$
 (B) $5 \cdot \sqrt{x^2+2x+5} - \ln|x+1-\sqrt{x^2+2x+5}| + c$
 (C) $5 \cdot \sqrt{x^2+2x+5} - \ln|x+1+\sqrt{x^2+2x+5}| + c$
 (D) $5 \cdot \sqrt{x^2+2x+5} + \ln|x+1+\sqrt{x^2+2x+5}| + c$

Q6 $\int \frac{x+3}{\sqrt{4x^2+4x-3}} dx = \frac{1}{4} \sqrt{f(x)} + \frac{5}{2} \int \frac{dx}{\sqrt{4x^2+4x-3}},$
 then find the minimum value of $f(x)$.

Q7 If $\int \frac{\cos \theta \cdot d\theta}{5+7 \sin \theta - 2 \cos^2 \theta} d\theta = A \log |B(\theta)| + c,$
 then $\frac{B(\theta)}{A}$ can be:

- (A) $\frac{2 \sin \theta + 1}{5(\sin \theta + 3)}$
 (B) $\frac{2 \sin \theta + 1}{\sin \theta + 3}$
 (C) $\frac{5(2 \sin \theta + 1)}{\sin \theta + 3}$
 (D) $\frac{5 \sin \theta}{2 \sin \theta + 1}$

Q8 If

$$\int \frac{2e^x + 3e^{-x}}{4e^x + 7e^{-x}} dx = \frac{1}{14} (ux + v \ln |4e^x + 7e^{-x}|) + c$$

Then $u + v$ is equal to:

Q9 $\int \frac{x^2+1}{x^4-x^2+1} dx$ is equal to

- (A) $\tan^{-1}\left(\frac{x^2-1}{x}\right) + c$
 (B) $\tan^{-1}\left(\frac{x^2+1}{2x}\right) + c$
 (C) $-\tan^{-1}\left(\frac{x^2-1}{x}\right) + c$
 (D) $\tan^{-1}\left(\frac{x^2-1}{2x}\right) + c$

Q10 $\int \frac{(4x+3)}{3x^2+3x+1} dx$

- (A) $\frac{1}{3} \ln(3x^2+3x+1) + \frac{2}{\sqrt{3}} \tan^{-1}(2\sqrt{3}x + \sqrt{3}) + C$
 (B) $\frac{2}{3} \ln(3x^2+3x+1) + \frac{2}{\sqrt{3}} \tan^{-1}(2\sqrt{3}x + \sqrt{3}) + C$
 (C) $\frac{2}{3} \ln(3x^2+3x+1) + \frac{1}{\sqrt{3}} \tan^{-1}(2\sqrt{3}x + \sqrt{3}) + C$



$$(D) \frac{2}{3} \ln(3x^2 + 3x + 1) - \frac{2}{\sqrt{3}} \tan^{-1}(2\sqrt{3}x + \sqrt{3}) + C$$

Q11 $\int \frac{x dx}{x^4 + x^2 + 1}$

(A) $\frac{2}{\sqrt{3}} \tan^{-1}\left(x^2 + \frac{1}{2}\right) + C$
 (B) $\frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2x^2 - 1}{\sqrt{3}}\right) + C$
 (C) $\frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2x^2 + 1}{\sqrt{3}}\right) + C$
 (D) $-\frac{1}{\sqrt{3}} \tan^{-1}\left(x^2 + \frac{1}{2}\right) + C$

Q12 $\int \frac{e^x dx}{\sqrt{5 - 4e^x + e^{2x}}}$

(A) $\ln(e^x - 2 + \sqrt{5 - 4e^x + e^{2x}}) + c$
 (B) $-\ln(e^x - 2 + \sqrt{5 - 4e^x + e^{2x}}) + c$
 (C) $\ln(e^x + 2 + \sqrt{5 - 4e^x + e^{2x}}) + c$
 (D) $\ln(e^x - 2 - \sqrt{5 - 4e^x + e^{2x}}) + c$

Q13 $\int x \cdot 2^{\ln(x^2+1)} dx$ equals

(A) $\frac{2^{\ln(x^2+1)}}{2(\ln 2 + 1)} + C$
 (B) $\frac{(x^2+1)2^{\ln(x^2+1)}}{2(\ln 2 + 1)} + C$
 (C) $\frac{(x^2+1)^{\ln 2}}{2(\ln 2 + 1)} + C$
 (D) $\frac{1}{8}(\ln 6x^2)^3 + C$

Q14 $\int \operatorname{cosec}^4 x dx$ is equal to

(A) $\cot x + \frac{\cot^3 x}{3} + C$
 (B) $\tan x + \frac{\tan^3 x}{3} + C$
 (C) $-\cot x - \frac{\cot^3 x}{3} + C$
 (D) $-\tan x - \frac{\tan^3 x}{3} + C$

Q15 $\int \frac{dx}{2 + \sin^2 x}$

(A) $\frac{1}{\sqrt{6}} \tan^{-1}\left(\frac{\sqrt{3} \tan x}{\sqrt{2}}\right) + c$
 (B) $\frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3} \tan x}{\sqrt{2}}\right) + c$
 (C) $-\frac{1}{\sqrt{6}} \tan^{-1}\left(\frac{\sqrt{3} \tan x}{\sqrt{2}}\right) + c$
 (D) $-\frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3} \tan x}{\sqrt{2}}\right) + c$

Q16 $\int \frac{\sin 2x}{\sin^2 x + 2 \cos^2 x} dx$ is equal to

(A) $-\log(1 + \sin^2 x) + C$
 (B) $\log(1 + \cos^2 x) + C$
 (C) $-\log(1 + \cos^2 x) + C$
 (D) $\log(1 + \tan^2 x) + C$

Q17 $\int \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$

(A) $\cot^{-1}(\tan^2 x) + L$
 (B) $\tan^{-1}(\tan^2 x) + c$
 (C) $\cot^{-1}(\cot^2 x) + c$
 (D) $\tan^{-1}(\cot^2 x) + c$

Q18 $\int \frac{\sin x + 3 \cos x + 1}{\sin x - 3 \cos x - 1} dx$

(A) $-\frac{4}{5}x + \frac{3}{5} \ln |\sin x - 3 \cos x - 1|$
 $+ \frac{1}{15} \ln \left| \frac{\tan(x/2) - 1}{\tan(x/2) + 2} \right| + c$
 (B) $\frac{4}{5}x + \frac{3}{5} \ln |\sin x - 3 \cos x - 1|$
 $+ \frac{1}{15} \ln \left| \frac{\tan(x/2) - 1}{\tan(x/2) + 2} \right| + c$
 (C) $-\frac{4}{5}x + \frac{3}{5} \ln |\sin x - 3 \cos x - 1|$
 $- \frac{1}{15} \ln \left| \frac{\tan(x/2) - 1}{\tan(x/2) + 2} \right| + c$
 (D) $\frac{4}{5}x + \frac{3}{5} \ln |\sin x - 3 \cos x - 1|$
 $- \frac{1}{15} \ln \left| \frac{\tan(x/2) - 1}{\tan(x/2) + 2} \right| + c$

Q19 $\int \frac{dx}{3 \sin^2 x + \sin x \cos x + 1}$

(A) $-\frac{2}{\sqrt{15}} \tan^{-1}\left(\frac{8 \tan x + 1}{\sqrt{15}}\right) + C$
 (B) $\frac{2}{\sqrt{15}} \tan^{-1}\left(\frac{8 \tan x - 1}{\sqrt{15}}\right) + C$
 (C) $\frac{1}{\sqrt{15}} \tan^{-1}\left(\frac{8 \tan x + 1}{\sqrt{15}}\right) + C$
 (D) $\frac{2}{\sqrt{15}} \tan^{-1}\left(\frac{8 \tan x + 1}{\sqrt{15}}\right) + C$

Q20 $\int \frac{x^2 + x - 6}{(x-2)(x-1)} dx$ is equal to:

(A) $x + 2 \log(x-1) + c$
 (B) $2x + 2 \log(x-1) + c$



(C) $x + 4 \log(1 - x) + c$

(D) $x + 4 \log(x - 1) + c$

Q21 $\int \frac{(x^2-1)}{(x^4+3x^2+1) \cdot \tan^{-1}\left(x+\frac{1}{x}\right)} dx$ is equal to:

(A) $\tan^{-1}\left(x + \frac{1}{x}\right) + c$

(B) $\cot^{-1}\left(x + \frac{1}{x}\right) + c$

(C) $\log\left(x + \frac{1}{x}\right) + c$

(D) $\log\left[\tan^{-1}\left(x + \frac{1}{x}\right)\right] + c$



Answer Key

- Q1 $\frac{1}{4}\sin^{-1}\left(\frac{4x}{3}\right) + c$
Q2 $\frac{1}{10}\tan^{-1}\left(\frac{2x}{5}\right) + c$
Q3 (C)
Q4 (B)
Q5 (C)
Q6 -4
Q7 (C)
Q8 7
Q9 (A)
Q10 (B)

- Q11 (C)
Q12 (A)
Q13 (B)
Q14 (C)
Q15 (A)
Q16 (C)
Q17 (B)
Q18 (A)
Q19 (D)
Q20 (D)
Q21 (D)



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